

Unit Functional Design Specification – Coal Boiler

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Following table describes the EM classes that shall be used along with their instance configuration.

EM Class	Sequence Name
EM_BOILER1_SBC	Soot Blower Control 1
EM_BOILER1_BPC	Bin Pump Control 1
EM_BOILER2_SBC	Soot Blower Control 2
EM_BOILER2_BPC	Bin Pump Control 1

1 EQUIPMENT MODULE CLASS DESIGN – EM_BOILER1_SBC

1.1 Functional Description

EM_BOILER1_SBC is a class based equipment module that perform soot blower control operation. EM_BOILER1_SBC is derived from the Command Driven equipment module class _EM_CMD32_130 which is defined in a separate document. Refer Functional Design Specification for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Soot Blower Control Command and It shall be released in same command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

1.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions.

Fail Monitor Name	Condition
HM01	Instance Devices Ownership Lost.
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.
SM11	Device 11 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Device 14 Failed.
SM15	Device 15 Failed.
SM16	Device 16 Failed.
SM17	Device 17 Failed.
SM18	Device 18 Failed.
SM19	Device 19 Failed.
SM20	Device 20 Failed.
SM21	Device 21 Failed.
SM22	Device 22 Failed.

Table 1: Hold Monitors

1.3 Sequences

This equipment module class has multiple states that can be commanded to set up specific Boiler paths. Table 2 below shows the control module states for each equipment module control state.

Control Module	Hold	Soot Blower
DEV1	X	O
DEV2	X	O
DEV3	X	O
DEV4	X	X
DEV5	X	O
DEV7	X	O
DEV8	X	O
DEV9	X	X
DEV10	X	O
DEV12	X	O
DEV13	O	O
DEV14	O	O
DEV15	O	O
DEV16	O	O
DEV17	O	O
DEV18	O	O
DEV19	O	O
DEV20	O	O
DEV21	O	O
DEV22	O	O

Table 2: Sequence Commands

O – Device Open or Start

X – Device Close or Stop

1.4 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	_EM_BOILER1_SBC	EM_BOILER1_SBC
DEV01	_DO_130_SE	BL1_V1010
DEV02	_DO_130_SE	BL1_FN1010A
DEV03	_DO_130_SE	BL1_V1010A2
DEV04	_DO_130_SE	BL1_V1010A1
DEV05	_DO_130_SE	BL1_IGI1010A
DEV07	_DO_130_SE	BL1_FN1010B
DEV08	_DO_130_SE	BL1_V1010B2
DEV09	_DO_130_SE	BL1_V1010B1
DEV10	_DO_130_SE	BL1_IGI1010B
DEV12	_DO_130_SE	BL1-FN1010*
DEV13	_DO_130_SE	BL1_V1010A4
DEV14	_DO_130_SE	BL1_V1010B4
DEV15	_DO_130_SE	BL1_V1010C4
DEV16	_DO_130_SE	BL1_V1010D4
DEV17	_DO_130_SE	BL1_V1010E4
DEV18	_DO_130_SE	BL1_V1010F4
DEV19	_DO_130_SE	BL1_V1010G4
DEV20	_DO_130_SE	BL1_V1010H4
DEV21	_DO_130_SE	BL1_V1010I4
DEV22	_DO_130_SE	BL1_V1010J4

Table 3: Instances

1.5 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

1.5.1 Command 00 – Hold

Step / Transition No.	Step / Transition Name	Functional Description
S0000	Start	Msg : Starting Hold Command
T0010	Hold	Go to Step S0010

S0010	Hold	Close DEV1,DEV2,DEV3, DEV4,DEV5,DEV7,DEV8, DEV9,DEV10,DEV12,DEV13,DEV14,DEV15,DEV16,DEV17,DEV18,DEV19,DEV20,DEV21,DEV22
T9900	Reset	Go to Step S9900
S9900	Reset	Reset flags and disable failure monitors. Enable Hold Req for DEV1,DEV2,DEV3,DEV4,DEV5,DEV7,DEV8,DEV9,DEV10,DEV12,DEV13,DEV14,DEV15,DEV16,DEV17,DEV18,DEV19,DEV20,DEV21 AND DEV22
END	Command Complete	True

Table 4: Hold Sequence

1.5.2 Command 01 – Soot Blower

Step / Transition No.	Step / Transition Name	Functional Description
S0000	Start	Msg : Starting Soot Blower Command
T0010	Initializations	Go to Step S0010
S0010	Initializations	Initialize Msg, Timers and Counters Acquire Device (DEV01, DEV02, DEV03, DEV04, DEV05, DEV07, DEV08, DEV09, DEV10, DEV12, DEV13, DEV14, DEV15, DEV16, DEV17, DEV18, DEV19, DEV20, DEV21 and DEV22)
T0020A	Acquire Complete	If Devices are acquired within 20 seconds then go to Step S0030
T0020B	Acquire Failed	If Devices are not acquired within 20 seconds then go to Step S0020
S0020	Acquire Failed	Operator Prompt : Acquisition failed for "Device Names"
T0040	Operator Ack	If Operator Clicks OK then return to Step S0010
S0030	Enable HM And SM	Enable Hold Monitors (HM01,) Enable Sentinel Monitors (SM01, SM02, SM03, SM04, SM05, SM07, SM08, SM09, SM10, SM12, SM13, SM14, SM15, SM16, SM17, SM18, SM19, SM20, SM21 and SM22)
T0050	Wait	Go to Step S0050

Step / Transition No.	Step / Transition Name	Functional Description
S0050	Start	Open DEV01 and DEV02
		Start TP01 Timer
T0060	Wait	If TP01 Timer complete then go to Step S0060
S0060	Open V1010A2	Open DEV3
		Start TP02 Timer
T0070	Wait	If TP02 Timer complete then go to Step S0070
S0070	Close V1010A2 and V1010A1	Close DEV3 and DEV4
T0080	Wait	Delay 0.5 second, then go to Step S0080
S0080	Open IGI1010A	Open DEV5
T0090	Wait	Delay 2 second, then go to Step S0090
S0090	Open V1010A1	Open DEV4
		Start TP05
T0100A	Wait	Judge blowing times reach set value or not, If TP05 count 3 times then go to Step S0200
T0100B	Wait	If no go to Step S0060
S0200	Open FN1010B	Open DEV7
		Start TP03 Timer
T0300	Wait	If TP03 Timer complete then go to Step S0300
S0300	Open V1010B2	Open DEV8
		Start TP04 Timer
T0400	Wait	If TP04 Timer complete then go to Step S0400
S400	Close V1010B2 and V1010B1	Close DEV8 and DEV9
T0500	Wait	Delay 0.5 seconds then go to step S0500
S0500	Open IGI1010B	Open DEV10

Step / Transition No.	Step / Transition Name	Functional Description
T0600	Wait	Delay 2 seconds then go to step S0600
S0600	Open V1010B1	Open DEV9
		Start TP06s
T0700A	Wait	Judge blowing times reach set value or not. If TP06 count 3 times then go to Step S0800
T0700B	Wait	If no go to Step S0300
S0800	Open FN1010*	Open DEV12
T0900	Wait	Go to step S0900
S0900	Judge	Judge blowing is complete (total 10 lines)
T1000	Wait	If No go te step S0050, If Yes go to step S1000
S1000	Stop sootblowing	Close DEV1, Open DEV13, DEV14, DEV15, DEV16, DEV17, DEV18, DEV19, DEV20, DEV21, DEV22
T2000	Complete	Go to step S9900
S9900	Complete	Msg: Soot Blower Complete
END	Command Complete	True

Table 5: Soot Blower Sequence

1.6 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Default Value	Units	Available on Detail Display
Operating Parameters				
N/A				
Tuning Parameters				
TP01	1 st Blowing time	25	Sec	
TP02	1 st Inflation time	5	Sec	
TP03	2 nd Blowing time	25	Sec	
TP04	2 nd Inflation time	5	Sec	
TP05	1 st Number of time	3	N/A	
TP06	2 nd Number of time	3	N/A	

Table 6: Operating and Tuning Parameters

1.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_BOILER1_SBC	BOILER 1: Soot Blower Control EM

Table 7: Named Set Details

Details of states for NS_BOILER1_SBC named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Soot Blower

Table 8: Name Set State Details

2 EQUIPMENT MODULE CLASS DESIGN – EM_BOILER1_BPC

2.1 Functional Description

EM_BOILER1_BPC is a class based equipment module that perform Bin Pump control operation. EM_BOILER1_BPC is derived from the Command Driven equipment module class _EM_CMD16_130 which is defined in a separate document. Refer Functional Design Specification for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Bin Pump Control Command and It shall be released in same command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

2.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions.

Fail Monitor Name	Condition
HM01	Instance Devices Ownership Lost.
HM02	Remote Control Signal Ok
HM04	Device 1 Running
HM05	Device 7 $\geq$ 0.5 MPa
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.

Table 9: Hold Monitors

2.3 Sequences

This equipment module class has multiple states that can be commanded to set up specific boiler paths. Table 10 below shows the control module states for each equipment module control state.

Control Module	Hold	Bin Pump
DEV01	X	O
DEV02	X	O
DEV03	X	O
DEV04	X	O
DEV05	X	O
DEV06	X	O

Table 10: Sequence Commands

O – Device Open or Start

X – Device Close or Stop

2.4 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	_EM_BOILER1_BPC	EM_BOILER1_BPC
DEV01		BL0_DC0070
DEV02	_DO_130_SE	BL1_V1071C
DEV03	_DO_130_SE	BL1_V1070
DEV04	_DO_130_SE	BL1_V1071D
DEV05	_DO_130_SE	BL1_V1071B
DEV06	_DO_130_SE	BL1_V1071A
DEV07	_AI_130_SE	BL1_PT1070
DEV08	_AI_130_SE	BL1_PT1071

Table 11: Instances

2.5 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

2.5.1 Command 00 – Hold

Step / Transition No.	Step / Transition Name	Functional Description
-----------------------	------------------------	------------------------

S0000	Start	Msg : Starting Hold Command
T0010	Hold	Go to Step S0010
S0010	Hold	Stop DEV01
		Close DEV02, DEV03, DEV04, DEV05 and DEV06
T9900	Reset	Go to Step S9900
S9900	Reset	Reset flags and disable failure monitors.
		Enable Hold Req for DEV01, DEV02, DEV03, DEV04, DEV05 and DEV06
END	Command Complete	True

Table 12: Hold Sequence

2.5.2

Command 01 – Bin Pump Control

Step / Transition No.	Step / Transition Name	Functional Description
S0000	Start	Msg : Starting Bin Pump Command
T0010	Initializations	Go to Step S0010
S0010	Initializations	Initialize Msg, Timers and Counters
		Acquire Device (DEV01, DEV02, DEV03, DEV04, DEV05 and DEV06)
T0020A	Acquire Complete	If Devices are acquired within 20 seconds then go to Step S0030
T0020B	Acquire Failed	If Devices are not acquired within 20 seconds then go to Step S0020
S0020	Acquire Failed	Operator Prompt : Acquisition failed for "Device Names"
T0040	Operator Ack	If Operator Clicks OK then return to Step S0010
S0030	Enable HM And SM	Enable Hold Monitors (HM01, HM02, HM03 and HM04)
		Enable Sentinel Monitors (SM01, SM02, SM03, SM04, SM05 and SM06)
T0050	Wait	Go to Step S0050

Step / Transition No.	Step / Transition Name	Functional Description
S0050	Start	Open DEV02
T0070	Wait	If DEV8 $\geq$ 0.15 MPa and delay 3 seconds then go to Step S0060
S0060	Anti-clogging valve	Open DEV03
T0080	Wait	Delay 3 seconds then go to Step S0070
S0070	Delivery valve	Open DEV04
T0090	Wait	If DEV8 $\leq$ 0.05 MPa then go to step S0080
S0080	Close	Close DEV2 and DEV3
T0100	Wait	Delay 6 seconds then go to Step S0090
S0090	Delivery valve	Close DEV04
		There is stop signal then stop process
T0200	Wait	If no man signal, no stop signal and delay 3 seconds then go to step S0100. If stop signal then go to step S0050
S0100	Exhaust valve	Open DEV05
T0300	Wait	Delay 2 seconds then go to step S0200
S0200	Material feeding valve	Open DEV06
		Start T1 timer
T0400A	Wait	If T1 Timer complete then go to Step S0300
T0400B	Wait	There is man or stop signal detected then stop DEV05,DEV06 and stop process
S0300	Close valve	Close DEV05 and DEV06
T0500	Wait	Go to step S0050

Table 13: Filter Press Closing Sequence

2.6 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Default Value	Units	Available on Detail Display
Operating Parameters				
N/A				
Tuning Parameters				
T1	Feeding Time	60	Sec	

Table 14: Operating and Tuning Parameters

2.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_BOILER1_BPC	Boiler 1: Bin Pump Control EM

Table 15: Named Set Details

Details of states for NS_BOILER1_BPC named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Bin Pump Control

Table 16: Name Set State Details

3 EQUIPMENT MODULE CLASS DESIGN – EM_BOILER2_SBC

3.1 Functional Description

EM_BOILER2_SBC is a class based equipment module that perform soot blower control operation. EM_BOILER2_SBC is derived from the Command Driven equipment module class _EM_CMD32_130 which is defined in a separate document. Refer Functional Design Specification for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Soot Blower Control Command and It shall be released in same command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

3.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions.

Fail Monitor Name	Condition
HM01	Instance Devices Ownership Lost.
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Device 14 Failed.
SM15	Device 15 Failed.
SM16	Device 16 Failed.
SM17	Device 17 Failed.
SM18	Device 18 Failed.
SM19	Device 19 Failed.
SM20	Device 20 Failed.
SM21	Device 21 Failed.
SM22	Device 22 Failed.

Table 17: Hold Monitors

3.3 Sequences

This equipment module class has multiple states that can be commanded to set up specific boiler paths.

Control Module	Hold	Soot Blower
DEV1	X	O
DEV2	X	O
DEV3	X	O
DEV4	X	X
DEV5	X	O
DEV7	X	O
DEV8	X	O
DEV9	X	X
DEV10	X	O
DEV12	X	O
DEV13	O	O
DEV14	O	O
DEV15	O	O
DEV16	O	O
DEV17	O	O
DEV18	O	O
DEV19	O	O
DEV20	O	O
DEV21	O	O
DEV22	O	O

Table 18: Sequence Commands

O – Device Open or Start

X – Device Close or Stop

3.4 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	_EM_BOILER2_SBC	EM_BOILER2_SBC
DEV01	_DO_130_SE	BL2_V2010
DEV02	_DO_130_SE	BL2_FN2010A
DEV03	_DO_130_SE	BL2_V2010A2
DEV04	_DO_130_SE	BL2_V2010A1
DEV05	_DO_130_SE	BL2_IGI2010A
DEV07	_DO_130_SE	BL2_FN2010B
DEV08	_DO_130_SE	BL2_V2010B2
DEV09	_DO_130_SE	BL2_V2010B1
DEV10	_DO_130_SE	BL2_IGI2010B
DEV12	_DO_130_SE	BL2_FN2010*
DEV13	_DO_130_SE	BL2_V2010A4
DEV14	_DO_130_SE	BL2_V2010B4
DEV15	_DO_130_SE	BL2_V2010C4
DEV16	_DO_130_SE	BL2_V2010D4
DEV17	_DO_130_SE	BL2_V2010E4
DEV18	_DO_130_SE	BL2_V2010F4
DEV19	_DO_130_SE	BL2_V2010G4
DEV20	_DO_130_SE	BL2_V2010H4
DEV21	_DO_130_SE	BL2_V2010I4
DEV22	_DO_130_SE	BL2_V2010J4

Table 19: Instances

3.5 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

3.5.1 Command 00 – Hold

Step / Transition No.	Step / Transition Name	Functional Description
S0000	Start	Msg : Starting Hold Command
T0010	Hold	Go to Step S0010
S0010	Hold	Close DEV1,DEV2,DEV3,DEV4,DEV5,DEV7,DEV8,DEV9,DEV10,DEV12,DEV13,DEV14,DEV15,DEV16,DEV17,DEV18,DEV19,DEV20,DEV21,DEV22
T9900	Reset	Go to Step S9900
S9900	Reset	Reset flags and disable failure monitors.
		Enable Hold Req for DEV1,DEV2,DEV3,DEV4,DEV5,DEV7,DEV8,DEV9,DEV10,DEV12,DEV13,DEV14,DEV15,DEV16,DEV17,DEV18,DEV19,DEV20,DEV21 AND DEV22
END	Command Complete	True

Table 20: Hold Sequence

3.5.2 Command 01 – Soot Blower

Step / Transition No.	Step / Transition Name	Functional Description
S0000	Start	Msg : Starting Soot Blower Command
T0010	Initializations	Go to Step S0010
S0010	Initializations	Initialize Msg, Timers and Counters
		Acquire Device (DEV01, DEV02, DEV03, DEV04, DEV05, DEV07, DEV08, DEV09, DEV10, DEV12, DEV13, DEV14, DEV15, DEV16, DEV17, DEV18, DEV19, DEV20, DEV21 and DEV22)
T0020A	Acquire Complete	If Devices are aquired within 20 seconds then go to Step S0030
T0020B	Acquire Failed	If Devices are not aquired within 20 seconds then go to Step S0020
S0020	Acquire Failed	Operator Prompt : Aquisition failed for “Device Names”
T0040	Operator Ack	If Operator Clicks OK then return to Step S0010
S0030	Enable HM And SM	Enable Hold Monitors (HM01,)
		Enable Sentinel Monitors (SM01, SM02, SM03, SM04, SM05, SM07, SM08, SM09, SM10, SM12, SM13, SM14, SM15, SM16, SM17, SM18, SM19, SM20, SM21 and SM22)
T0050	Wait	Go to Step S0050
S0050	Start	Open DEV01 and DEV02
		Start TP01 Timer
T0060	Wait	If TP01 Timer complete then go to Step S0060
S0060	Open V2010A2	Open DEV3
		Start TP02 Timer
T0070	Wait	If TP02 Timer complete then go to Step S0070
S0070	Close V2010A2 and V2010A1	Close DEV3 and DEV4
T0080	Wait	Delay 0.5 second, then go to Step S0080

Step / Transition No.	Step / Transition Name	Functional Description
S0080	Open IGI2010A	Open DEV5
T0090	Wait	Delay 2 second, then go to Step S0090
S0090	Open V2010A1	Open DEV4
		Start TP05
T0100A	Wait	Judge blowing times reach set value or not, If TP05 count 3 times then go to Step S0200
T0100B	Wait	If no go to Step S0060
S0200	Open FN2010B	Open DEV7
		Start TP03 Timer
T0300	Wait	If TP03 Timer complete then go to Step S0300
S0300	Open V2010B2	Open DEV8
		Start TP04 Timer
T0400	Wait	If TP04 Timer complete then go to Step S0400
S400	Close V2010B2 and V2010B1	Close DEV8 and DEV9
T0500	Wait	Delay 0.5 seconds then go to step S0500
S0500	Open IGI2010B	Open DEV10
T0600	Wait	Delay 2 seconds then go to step S0600
S0600	Open V2010B1	Open DEV9
		Start TP06
T0700A	Wait	Judge blowing times reach set value or not, If TP06 count 3 times then go to Step S0800
T0700B	Wait	If no go to Step S0300
S0800	Open FN2010*	Open DEV12
T0900	Wait	Go to step S0900
S0900	Judge	Judge blowing is complete (total 10 lines)

Step / Transition No.	Step / Transition Name	Functional Description
T1000	Wait	If No go to step S0050, If Yes go to step S1000
S1000	Stop sootblowing	Close DEV1, Open DEV13, DEV14, DEV15, DEV16, DEV17, DEV18, DEV19, DEV20, DEV21, DEV22
T2000	Complete	Go to step S9900
S9900	Complete	Msg: Soot Blower Complete
END	Command Complete	True

Table 21: Soot Blower Sequence

3.6 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Default Value	Units	Available on Detail Display
Operating Parameters				
N/A				
Tuning Parameters				
TP01	1 st Blowing time	25	Sec	
TP02	1 st Inflation time	5	Sec	
TP03	2 nd Blowing time	25	Sec	
TP04	2 nd Inflation time	5	Sec	
TP05	1 st Number of time	3	N/A	
TP06	2 nd Number of time	3	N/A	

Table 22: Operating and Tuning Parameters

3.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_BOILER2_SBC	Boiler 2: Soot Blower Control EM

Table 23: Named Set Details

Details of states for NS_BOILER2_SBC named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Soot Blower

Table 24: Name Set State Details

4 EQUIPMENT MODULE CLASS DESIGN – EM_BOILER2_BPC

4.1 Functional Description

EM_BOILER2_BPC is a class based equipment module that perform Bin Pump control operation. EM_BOILER2_BPC is derived from the Command Driven equipment module class _EM_CMD16_130 which is defined in a separate document. Refer Functional Design Specification for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Bin Pump Control Command and It shall be released in same command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

4.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions.

Fail Monitor Name	Condition
HM01	Instance Devices Ownership Lost.
HM02	Remote Control Signal Ok
HM03	Device 1 Running
HM04	Device7 $\geq$ 0.5 MPa
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.

Table 25: Hold Monitors

4.3 Sequences

This equipment module class has multiple states that can be commanded to set up specific boiler paths. Table 10 below shows the control module states for each equipment module control state.

Control Module	Hold	Bin Pump
DEV01	X	O
DEV02	X	O
DEV03	X	O
DEV04	X	O
DEV05	X	O
DEV06	X	O

Table 26: Sequence Commands

O – Device Open or Start

X – Device Close or Stop

4.4 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	_EM_BOILER1_BPC	EM_BOILER1_BPC
DEV01		BL0_DC0070
DEV02	_DO_130_SE	BL2_V2071C
DEV03	_DO_130_SE	BL2_V2070
DEV04	_DO_130_SE	BL2_V2071D
DEV05	_DO_130_SE	BL2_V2071B
DEV06	_DO_130_SE	BL2_V2071A
DEV07	_AI_130_SE	BL2_PT2070
DEV08	_AI_130_SE	BL2_PT2071

Table 27: Instances

4.5 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

4.5.1 Command 00 – Hold

Step / Transition No.	Step / Transition Name	Functional Description
-----------------------	------------------------	------------------------

S0000	Start	Msg : Starting Hold Command
T0010	Hold	Go to Step S0010
S0010	Hold	Stop DEV01
		Close DEV02, DEV03, DEV04, DEV05 and DEV06
T9900	Reset	Go to Step S9900
S9900	Reset	Reset flags and disable failure monitors.
		Enable Hold Req for DEV01, DEV02, DEV03, DEV04, DEV05 and DEV06
END	Command Complete	True

Table 28: Hold Sequence

4.5.2 Command 01 – Bin Pump Control

Step / Transition No.	Step / Transition Name	Functional Description
S0000	Start	Msg : Starting Bin Pump Command
T0010	Initializations	Go to Step S0010
S0010	Initializations	Initialize Msg, Timers and Counters
		Acquire Device (DEV01, DEV02, DEV03, DEV04, DEV05 and DEV06)
T0020A	Acquire Complete	If Devices are acquired within 20 seconds then go to Step S0030
T0020B	Acquire Failed	If Devices are not acquired within 20 seconds then go to Step S0020
S0020	Acquire Failed	Operator Prompt : Acquisition failed for "Device Names"
T0040	Operator Ack	If Operator Clicks OK then return to Step S0010
S0030	Enable HM And SM	Enable Hold Monitors (HM01, HM02, HM03 and HM04)
		Enable Sentinel Monitors (SM01, SM02, SM03, SM04, SM05 and SM06)
T0050	Wait	Go to Step S0050
S0050	Start	Open DEV02

Step / Transition No.	Step / Transition Name	Functional Description
T0070	Wait	If DEV08 $\geq$ 0.15 MPa and delay 3 seconds then go to Step S0060
S0060	Anti-clogging valve	Open DEV03
T0080	Wait	Delay 3 seconds then go to Step S0070
S0070	Delivery valve	Open DEV04
T0090	Wait	If DEV08 $\leq$ 0.05 MPa then go to step S0080
S0080	Close	Close DEV2 and DEV3
T0100	Wait	Delay 6 seconds then go to Step S0090
S0090	Delivery valve	Close DEV04
		There is stop signal then stop process
T0200	Wait	If no man signal, no stop signal and delay 3 seconds then go to step S0100. If stop signal then go to step S0050
S0100	Exhaust valve	Open DEV05
T0300	Wait	Delay 2 seconds then go to step S0200
S0200	Material feeding valve	Open DEV06
		Start T1 timer
T0400A	Wait	There is man or stop signal detected then stop DEV05,DEV06 and stop process
T0400B	Wait	If T1 Timer complete then go to Step S0300
S0300	Close valve	Close DEV05 and DEV06
T0500	Wait	Go to step S0050

Table 29: Filter Press Closing Sequence

4.6 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Default Value	Units	Available on Detail Display
Operating Parameters				
N/A				
Tuning Parameters				
T1	Feeding Time	60	Sec	

Table 30: Operating and Tuning Parameters

4.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_BOILER2_BPC	Boiler 1: Bin Pump Control EM

Table 31: Named Set Details

Details of states for NS_BOILER2_BPC named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Bin Pump Control

Table 32: Name Set State Details

5 MODULE INTERLOCKS

5.1 Functional Description

The section below explains the interlock configurations

Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_FN1016	Condition 1	Bed layer pressure difference (left)	LdR1032A HH: 9.6 Kpa, Start FN1016	Start	NO	NO	NO	NO	NO
BL1_FN1016	Condition 2	Bed layer pressure difference (Right)	LdR1032B HH: 9.6 Kpa, Start FN1016	Start	NO	NO	NO	NO	NO
BL1_SC1001	Condition 2	Slag cooler cooling fan	FN1016 Stop, SC1001 Stop	Stop	NO	NO	NO	NO	NO
BL1_FN1016	Condition 3	Bed layer pressure difference (left)	LdR1032A LL $\leq$ 8.9 Kpa, Stop FN1016	Stop	NO	NO	NO	NO	NO
BL1_FN1016	Condition 4	Bed layer pressure difference (Right)	LdR1032B LL $\leq$ 8.9 Kpa, Stop FN1016	Stop	NO	NO	NO	NO	NO
BL1_FN1016	Condition 5	Slag cooler cooling water flow low	FSL1025 = 1, Stop FN1016	Stop	NO	NO	NO	NO	NO
BL1_FN1016	Condition 6	Slag cooler cooling water temperature	TT1025 $\geq$ 75, Stop FN1016	Stop	NO	NO	NO	NO	NO
BL1_INT1	Condition 1	Induced draft fan trip or stops	FN1003 trip or stop, Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_INT1	Condition 2	Primary air fan trip or stops	FN1001 trip or stop, Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	NO	NO
BL1_INT1	Condition 3	Secondary air fan trip or stops	FN1002 trip or stop, Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	NO	NO
BL1_INT1	Condition 4	Manual stop boiler	Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	NO	NO
BL1_INT1	Condition 5	Furnace flue gas pressure HH (2 OF 2)	PT1033A/PT1033B HH: 900 Pa, Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT1	Condition 6	Furnace flue gas pressure LL (2 OF 2)	PT1033A/PT1033B LL: -900 Pa, Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT1	Condition 7	Drum level LL (2 OF 3)	LT1020A/LT1020B/LIS1020 LL: -150 mm, Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT1	Condition 8	Drum pressure HH (2 OF 2)	PT1020/PSH1020A HH: 1.9 Mpa, Stop FN1001, FN1002, FN1003, BC1001, BC1002, BC1004, RF231	Stop	NO	NO	NO	3 -5 s	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_INT2	Condition 1	Bed material temperature HH (3 OF 6)	TR1032A/TR1032B/TR1032C/ TR1032D/TR1032E/TR1032F HH: 1050, Stop BC1001, BC1002, RF231	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT2	Condition 2	Bed material temperature LL (3 OF 6)	TR1032A/TR1032B/TR1032C/ TR1032D/TR1032E/TR1032F LL: 700, Stop BC1001, BC1002, RF231	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT2	Condition 3	Oxygen content at flue LL	AT1038 LL: 3%, Stop BC1001, BC1002, RF231	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT3	Condition 1	#1 Belt coal feeder blocking coal signal	LSH1051 Blocking coal signal, Stop BC1001 & FN1014	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT3	Condition 2	#1 Belt coal feeder anti-deviation device signal	BC1001_AD Anti-deviation device signal, Stop BC1001 & FN1014	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT3	Condition 3	#1 Outlet electric door fully close	MV1001_ZSC Fully close, Stop BC1001 & FN1014	Stop	NO	NO	NO	NO	NO
BL1_V1001	Condition 1	HH: 75 Close MV1001	TT1052 HH: 75 Close MV1001	Close	NO	NO	NO	3 -5 s	NO
BL1_INT4	Condition 1	#2 Belt coal feeder blocking coal signal	LSH1052 Blocking coal signal, Stop BC1002 & FN1015	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT4	Condition 2	#2 Belt coal feeder anti-deviation device signal	BC1002_AD Anti-deviation device signal, Stop BC1002 & FN1015	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT4	Condition 3	#2 Outlet electric door fully close	MV1002_ZSC Fully close, Stop BC1002 & FN1015	Stop	NO	NO	NO	NO	NO
BL1_V1002	Condition 1	HH: 75 Close MV1002	TT1052 HH: 75 Close MV1002	Close	NO	NO	NO	3 -5 s	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_INT5	Condition 1	HH: 80 Stop FN1001	TR1001A HH: 80, Stop FN1001, FN1011	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT5	Condition 2	HH: 80 Stop FN1001	TR1001B HH: 80, Stop FN1001, FN1011	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT6	Condition 1	HH: 80 Stop FN1002	TR1002A HH: 80, Stop FN1002, FN1012	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT6	Condition 2	HH: 80 Stop FN1002	TR1002B HH: 80, Stop FN1002, FN1012	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT7	Condition 1	HH: 80 Stop FN1003	TR1003A HH: 80, Stop FN1003, FN1013	Stop	NO	NO	NO	3 -5 s	NO
BL1_INT7	Condition 2	HH: 80 Stop FN1003	TR1003B HH: 80, Stop FN1003, FN1013	Stop	NO	NO	NO	3 -5 s	NO
BL1_BC1001A	Condition 1	#1 Cleaning motor broken chain signal	CM1001_BK broken chain signal, stop CM1001	Stop	NO	NO	NO	3 -5 s	NO
BL1_BC1001A	Condition 2	#2 Cleaning motor broken chain signal	CM1002_BK broken chain signal, stop CM1002	Stop	NO	NO	NO	3 -5 s	NO
BLO_PU0001	Condition 2	FN001 Trip or stop	FN001 trip or stop, stop PU001	Stop	NO	NO	NO	NO	NO
BLO_PU0002	Condition 2	FN002 Trip or stop	FN002 trip or stop, stop PU002	Stop	NO	NO	NO	NO	NO
BL1_V1020	Condition 1	Drum level HH	LT1020A/LT1020B/LIS1020 HH: +150mm, Open XV1020	Open	NO	NO	NO	NO	3 -5 s
BL1_V1020	Condition 2	Drum level H	LT1020A/LT1020B/LIS1020 H: +50mm, Close XV1020	Close	NO	NO	NO	NO	NO
BL1_PCV1024	Condition 1	Drum pressure HH	PT1020/PSH1020A HH: 1.9 Mpa, Open PCV1024 (CV=100%)	Open	NO	NO	NO	NO	1 -2 s

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_PCV1024	Condition 2	Drum pressure	PT1020/PSH1020A : 1.6 Mpa, Close PCV1024 (CV=0%)	Close	NO	NO	NO	NO	NO
BL1_HE1061A	Condition 1	Temperature low	TT1067 ≤ 90, Start HE1061	Start	NO	NO	NO	NO	NO
BL1_HE1061A	Condition 2	Temperature high	TT1067 ≥ 105, Stop HE1061	Stop	NO	NO	NO	NO	NO
BL1_HE1062A	Condition 1	Temperature low	TT1068 ≤ 90, Start HE1062	Start	NO	NO	NO	NO	NO
BL1_HE1062A	Condition 2	Temperature high	TT1068 ≥ 105, Stop HE1062	Stop	NO	NO	NO	NO	NO
BL1_HE1063A	Condition 1	Temperature low	TT1069 ≤ 90, Start HE1063	Start	NO	NO	NO	NO	NO
BL1_HE1063A	Condition 2	Temperature high	TT1069 ≥ 105, Stop HE1063	Stop	NO	NO	NO	NO	NO
BL1_HE1061C	Condition 1	Temperature low	TT1064 ≤ 90, Start SH1061	Start	NO	NO	NO	NO	NO
BL1_HE1061C	Condition 2	Temperature high	TT1064 ≥ 105, Stop SH1061	Stop	NO	NO	NO	NO	NO
BL1_HE1062C	Condition 1	Temperature low	TT1065 ≤ 90, Start SH1062	Start	NO	NO	NO	NO	NO
BL1_HE1062C	Condition 2	Temperature high	TT1065 ≥ 105, Stop SH1062	Stop	NO	NO	NO	NO	NO
BL1_HE1063C	Condition 1	Temperature low	TT1066 ≤ 90, Start SH1063	Start	NO	NO	NO	NO	NO
BL1_HE1063C	Condition 2	Temperature high	TT1066 ≥ 105, Stop SH1063	Stop	NO	NO	NO	NO	NO
BL1_HE1061B	Condition 1	Temperature low	TT1061 ≤ 90, Start IH1061	Start	NO	NO	NO	NO	NO
BL1_HE1061B	Condition 2	Temperature high	TT1061 ≥ 105, Stop IH1061	Stop	NO	NO	NO	NO	NO
BL1_HE1062B	Condition 1	Temperature low	TT1062 ≤ 90, Start IH1062	Start	NO	NO	NO	NO	NO
BL1_HE1062B	Condition 2	Temperature high	TT1062 ≥ 105, Stop IH1062	Stop	NO	NO	NO	NO	NO
BL1_HE1063B	Condition 1	Temperature low	TT1063 ≤ 90, Start IH1063	Start	NO	NO	NO	NO	NO
BL1_HE1063B	Condition 2	Temperature high	TT1063 ≥ 105, Stop IH1063	Stop	NO	NO	NO	NO	NO
BL1_VB1061A	Condition 1	Quiver time	1 min, Start AR1061	Start	NO	NO	NO	NO	NO
BL1_VB1061A	Condition 2	Circulate time	10 min, Stop AR1061	Stop	NO	NO	NO	NO	NO
BL1_VB1062A	Condition 1	Quiver time	3 min, Start AR1062	Start	NO	NO	NO	NO	NO
BL1_VB1062A	Condition 2	Circulate time	30 min, Stop AR1062	Stop	NO	NO	NO	NO	NO
BL1_VB1063A	Condition 1	Quiver time	5 min, Start AR1063	Start	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_VB1063A	Condition 2	Circulate time	50 min, Stop AR1063	Stop	NO	NO	NO	NO	NO
BL1_VB1061B	Condition 1	Quiver time	2 min, Start CR1061	Start	NO	NO	NO	NO	NO
BL1_VB1061B	Condition 2	Circulate time	10 min, Stop CR1061	Stop	NO	NO	NO	NO	NO
BL1_VB1062B	Condition 1	Quiver time	4 min, Start CR1062	Start	NO	NO	NO	NO	NO
BL1_VB1062B	Condition 2	Circulate time	10 min, Stop CR1062	Stop	NO	NO	NO	NO	NO
BL1_VB1063B	Condition 1	Quiver time	6 min, Start CR1063	Start	NO	NO	NO	NO	NO
BL1_VB1063B	Condition 2	Circulate time	30 min, Stop CR1063	Stop	NO	NO	NO	NO	NO
BL1_HT1061	Condition 1	Trip or stop or MFT signal	FN1001, FN1002, FN1003 Trip or stop or MFT signal, Stop HT1061	Stop	NO	NO	NO	NO	NO
BL1_HT1061	Condition 2	Ash hopper high level signal	LSH1061 high level signal	Stop	NO	NO	NO	10 min	NO
BL1_HT1062	Condition 1	Stop or MFT signal	FN1001, FN1002, FN1003 stop or MFT signal, Stop HT1062	Stop	NO	NO	NO	NO	NO
BL1_HT1062	Condition 2	Ash hopper high level signal	LSH1063 high level signal	Stop	NO	NO	NO	10 min	NO
BL1_HT1063	Condition 1	Stop or MFT signal	FN1001, FN1002, FN1003 stop or MFT signal, Stop HT1063	Stop	NO	NO	NO	NO	NO
BL1_HT1063	Condition 2	Ash hopper high level signal	LSH1063 high level signal	Stop	NO	NO	NO	10 min	NO
BL1_FN1003	Condition 1	Trip or stops	FN1013 Trip or stops, Stop FN1003	Stop	NO	NO	NO	NO	NO
BL1_FN1003	Condition 2	No MFT signal	No MFT signal, Start FN1003	Start	NO	NO	NO	NO	NO
BL1_FN1003	Condition 3	No TT1003A HH: 80	No TT1003A HH: 80, Start FN1003	Start	NO	NO	NO	NO	NO
BL1_FN1003	Condition 4	No TT1003B HH: 80	No TT1003B HH: 80, Start FN1003	Start	NO	NO	NO	NO	NO
BL1_FN1011	Condition 1	Trip or stops	FN1003 Trip or stops, Stop FN1011	Stop	NO	NO	NO	NO	NO
BL1_FN1001	Condition 1	Trip or stops	FN1011 Trip or stops, Stop FN1001	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_FN1001	Condition 2	No MFT signal	No MFT signal, Start FN1001	Start	NO	NO	NO	NO	NO
BL1_FN1001	Condition 3	No TT1001A HH: 80	No TT1001A HH: 80, Start FN1001	Start	NO	NO	NO	NO	NO
BL1_FN1001	Condition 4	No TT1001B HH: 80	No TT1001B HH: 80, Start FN1001	Start	NO	NO	NO	NO	NO
BL1_INT8	Condition 1	Trip or stops	FN1001 Trip or stops, Stop FN1012, FN1014, FN1015, FN1004	Stop	NO	NO	NO	NO	NO
BL1_INT8	Condition 2	MV1001_ZSO signal ok	MV1001_ZSO signal ok, Start FN1014	Start	NO	NO	NO	NO	NO
BL1_INT8	Condition 3	MV1001_ZSO signal ok	MV1002_ZSO signal ok, Start FN1014	Start	NO	NO	NO	NO	NO
BL1_FN1002	Condition 1	Trip or stops	FN1012 Trip or stops, Stop FN1002	Stop	NO	NO	NO	NO	NO
BL1_FN1002	Condition 2	No MFT signal	No MFT signal, Start FN1002	Start	NO	NO	NO	NO	NO
BL1_FN1002	Condition 3	No TT1002A HH: 80	No TT1002A HH: 80, Start FN1002	Start	NO	NO	NO	NO	NO
BL1_FN1002	Condition 4	No TT1002B HH: 80	No TT1002B HH: 80, Start FN1002	Start	NO	NO	NO	NO	NO
BL1_BC1001	Condition 1	Trip or stops	FN1014 Trip or stops, Stop BC1001	Stop	NO	NO	NO	NO	NO
BL1_BC1001	Condition 2	No MFT signal	No MFT signal, Start BC1001	Start	NO	NO	NO	NO	NO
BL1_BC1001	Condition 3	No LSH1051 signal	No LSH1051 signal, Start BC1001	Start	NO	NO	NO	NO	NO
BL1_BC1001	Condition 4	No BC1001_AD signal	No BC1001_AD signal, Start BC1001	Start	NO	NO	NO	NO	NO
BL1_BC1002	Condition 1	Trip or stops	FN1015 Trip or stops, Stop BC1002	Stop	NO	NO	NO	NO	NO
BL1_BC1002	Condition 2	No MFT signal	No MFT signal, Start BC1002	Start	NO	NO	NO	NO	NO
BL1_BC1002	Condition 3	No LSH1052 signal	No LSH1052 signal, Start BC1002	Start	NO	NO	NO	NO	NO
BL1_BC1002	Condition 4	No BC1002_AD signal	No BC1002_AD signal, Start BC1002	Start	NO	NO	NO	NO	NO
BL1_INT9	Condition 1	Trip or stops	BC1001 Trip or stops, Stop RF231	Stop	NO	NO	NO	NO	NO
BL1_INT9	Condition 2	Trip or stops	BC1002 Trip or stops, Stop RF231	Stop	NO	NO	NO	NO	NO
BL1_INT9	Condition 3	PT230D H 0.4 MPa	PT230D H 0.4 Mpa, Stop RF231	Stop	NO	NO	NO	15s	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL1_INT9	Condition 4	No MFT signal	No MFT signal, Start RF231	Start	NO	NO	NO	NO	NO
BL2_FN2016	Condition 1	Bed layer pressure difference (left)	LdR2032A HH $\geq$ 9.6 Kpa, Start FN2016	Start	NO	NO	NO	NO	NO
BL2_FN2016	Condition 2	Bed layer pressure difference (Right)	LdR2032B HH $\geq$ 9.6 Kpa, Start FN2016	Start	NO	NO	NO	NO	NO
BL2_SC2001	Condition 2	Slag cooler cooling fan	FN2016 Stop, SC2001 Stop	Stop	NO	NO	NO	NO	NO
BL2_FN2016	Condition 3	Bed layer pressure difference (left)	LdR2032A LL $\leq$ 8.9 Kpa, Stop FN2016	Stop	NO	NO	NO	NO	NO
BL2_FN2016	Condition 4	Bed layer pressure difference (Right)	LdR2032B LL $\leq$ 8.9 Kpa, Stop FN2016	Stop	NO	NO	NO	NO	NO
BL2_FN2016	Condition 5	Slag cooler cooling water flow low	FSL2025 = 1, Stop FN2016	Stop	NO	NO	NO	NO	NO
BL2_FN2016	Condition 6	Slag cooler cooling water temperature	TT2025 $\geq$ 75°, Stop FN2016	Stop	NO	NO	NO	NO	NO
BL2_INT1	Condition 1	Induced draft fan trip or stops	FN2003 trip or stop, Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232	Stop	NO	NO	NO	NO	NO
BL2_INT1	Condition 2	Primary air fan trip or stops	FN2001 trip or stop, Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232	Stop	NO	NO	NO	NO	NO
BL2_INT1	Condition 3	Secondary air fan trip or stops	FN2002 trip or stop, Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232	Stop	NO	NO	NO	NO	NO
BL2_INT1	Condition 4	Manual stop boiler	Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL2_INT1	Condition 5	Furnace flue gas pressure HH	PT2033A/PT2033B HH: 900 Pa, Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232 (2 OF 2)	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT1	Condition 6	Furnace flue gas pressure LL	PT2033A/PT2033B LL: -900 Pa, Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232 (2 OF 2)	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT1	Condition 7	Drum level LL	LT2020A/LT2020B/LIS2020 LL: -150 mm, Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232 (2 OF 3)	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT1	Condition 8	Drum pressure HH	PT2020/PSH2020A HH: 1.9 Mpa, Stop FN2001, FN2002, FN2003, BC2001, BC2002, BC2004, RF232 (2 OF 2)	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT2	Condition 1	Bed material temperature HH	TR2032A/TR2032B/TR2032C/TR2032D/TR2032E/TR2032F HH: 1050°, Stop BC2001, BC2002, RF232 (3 OF 6)	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT2	Condition 2	Bed material temperature LL	TR2032A/TR2032B/TR2032C/TR2032D/TR2032E/TR2032F LL: 700°, Stop BC2001, BC2002, RF232 (3 OF 6)	Stop	NO	NO	NO	3 -5 s	NO

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BL2_INT2	Condition 3	Oxygen content at flue LL	AT1038 LL: 3%, Stop BC2001, BC2002, RF232	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT3	Condition 1	#1 Belt coal feeder blocking coal signal	LSH2051 Blocking coal signal, Stop BC2001 & FN2014	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT3	Condition 2	#1 Belt coal feeder anti-deviation device signal	BC2001_AD Anti-deviation device signal, Stop BC2001 & FN2014	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT3	Condition 3	#1 Outlet electric door fully close	MV2001_ZSC Fully close, Stop BC2001 & FN2014	Stop	NO	NO	NO	NO	NO
BL2_V2001	Condition 1	HH: 75° Close MV2001	TT2052 HH: 75° Close MV2001	Close	NO	NO	NO	3 -5 s	NO
BL2_INT4	Condition 1	#2 Belt coal feeder blocking coal signal	LSH2052 Blocking coal signal, Stop BC2002 & FN2015	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT4	Condition 2	#2 Belt coal feeder anti-deviation device signal	BC2002_AD Anti-deviation device signal, Stop BC2002 & FN2015	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT4	Condition 3	#2 Outlet electric door fully close	MV1002_ZSC Fully close, Stop BC2002 & FN2015	Stop	NO	NO	NO	NO	NO
BL2_V2002	Condition 1	HH: 75° Close MV2002	TT2052 HH: 75° Close MV2002	Close	NO	NO	NO	3 -5 s	NO
BL2_INT5	Condition 1	HH: 80° Stop FN2001	TR2001A HH: 80°, Stop FN2001, FN2011	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT5	Condition 2	HH: 80° Stop FN2001	TR2001B HH: 80°, Stop FN2001, FN2011	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT6	Condition 1	HH: 80° Stop FN2002	TR2002A HH: 80°, Stop FN2002, FN2012	Stop	NO	NO	NO	3 -5 s	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL2_INT6	Condition 2	HH: 80° Stop FN2002	TR2002B HH: 80°, Stop FN2002, FN2012	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT7	Condition 1	HH: 80° Stop FN2003	TR2003A HH: 80°, Stop FN2003, FN2013	Stop	NO	NO	NO	3 -5 s	NO
BL2_INT7	Condition 2	HH: 80° Stop FN2003	TR2003B HH: 80°, Stop FN2003, FN2013	Stop	NO	NO	NO	3 -5 s	NO
BL2_BC2001A	Condition 1	#1 Cleaning motor broken chain signal	CM2001_BK broken chain signal, stop CM2001	Stop	NO	NO	NO	3 -5 s	NO
BL2_BC2001A	Condition 2	#2 Cleaning motor broken chain signal	CM2002_BK broken chain signal, stop CM2002	Stop	NO	NO	NO	3 -5 s	NO
BLO_PU003	Condition 2	FN003 Trip or stop	FN003 trip or stop, stop PU003	Stop	NO	NO	NO	NO	NO
BLO_PU004	Condition 2	FN004 Trip or stop	FN004 trip or stop, stop PU004	Stop	NO	NO	NO	NO	NO
BL2_V2020	Condition 1	Drum level HH	LT2020A/LT2020B/LIS2020 HH: +150mm	Open	NO	NO	NO	NO	3 -5 s
BL2_V2020	Condition 2	Drum level H	LT2020A/LT2020B/LIS2020 H: +50mm	Close	NO	NO	NO	NO	NO
BL2_PCV2024	Condition 1	Drum pressure HH	PT2020/PSH2020A HH: 1.9 Mpa, Open PCV2024 (CV=100%)	Open	NO	NO	NO	NO	1 -2 s
BL2_PCV2024	Condition 2	Drum pressure	PT2020/PSH2020A : 1.6 Mpa, Close PCV2024 (CV=0%)	Close	NO	NO	NO	NO	NO
BL2_HE2061A	Condition 1	Temperature low	TT2067 ≤ 90°, Start HE2061	Start	NO	NO	NO	NO	NO
BL2_HE2061A	Condition 2	Temperature high	TT2067 ≥ 105°, Stop HE2061	Stop	NO	NO	NO	NO	NO
BL2_HE2062A	Condition 1	Temperature low	TT2068 ≤ 90°, Start HE2062	Start	NO	NO	NO	NO	NO
BL2_HE1062A	Condition 2	Temperature high	TT1068 ≥ 105°, Stop HE2062	Stop	NO	NO	NO	NO	NO
BL2_HE2063A	Condition 1	Temperature low	TT2069 ≤ 90°, Start HE2063	Start	NO	NO	NO	NO	NO

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BL2_HE2063A	Condition 2	Temperature high	TT1069 $\geq$ 105°, Stop HE2063	Stop	NO	NO	NO	NO	NO
BL2_HE2061C	Condition 1	Temperature low	TT2064 $\leq$ 90°, Start SH2061	Start	NO	NO	NO	NO	NO
BL2_HE2061C	Condition 2	Temperature high	TT2064 $\geq$ 105°, Stop SH2061	Stop	NO	NO	NO	NO	NO
BL2_HE2062C	Condition 1	Temperature low	TT2065 $\leq$ 90°, Start SH2062	Start	NO	NO	NO	NO	NO
BL2_HE2062C	Condition 2	Temperature high	TT2065 $\geq$ 105°, Stop SH2062	Stop	NO	NO	NO	NO	NO
BL2_HE2063C	Condition 1	Temperature low	TT2066 $\leq$ 90°, Start SH2063	Start	NO	NO	NO	NO	NO
BL2_HE2063C	Condition 2	Temperature high	TT2066 $\geq$ 105°, Stop SH2063	Stop	NO	NO	NO	NO	NO
BL2_HE2061B	Condition 1	Temperature low	TT2061 $\leq$ 90°, Start IH2061	Start	NO	NO	NO	NO	NO
BL2_HE2061B	Condition 2	Temperature high	TT2061 $\geq$ 105°, Stop IH2061	Stop	NO	NO	NO	NO	NO
BL2_HE2062B	Condition 1	Temperature low	TT2062 $\leq$ 90°, Start IH2062	Start	NO	NO	NO	NO	NO
BL2_HE2062B	Condition 2	Temperature high	TT2062 $\geq$ 105°, Stop IH2062	Stop	NO	NO	NO	NO	NO
BL2_HE2063B	Condition 1	Temperature low	TT2063 $\leq$ 90°, Start IH2063	Start	NO	NO	NO	NO	NO
BL2_HE2063B	Condition 2	Temperature high	TT1063 $\geq$ 105°, Stop IH2063	Stop	NO	NO	NO	NO	NO
BL2_VB2061A	Condition 1	Quiver time	1min, Start AR2061	Start	NO	NO	NO	NO	NO
BL2_VB2061A	Condition 2	Circulate time	10min, Stop AR2061	Stop	NO	NO	NO	NO	NO
BL2_VB2062A	Condition 1	Quiver time	3min, Start AR2062	Start	NO	NO	NO	NO	NO
BL2_VB2062A	Condition 2	Circulate time	30min, Stop AR2062	Stop	NO	NO	NO	NO	NO
BL2_VB2063A	Condition 1	Quiver time	5min, Stop AR2063	Start	NO	NO	NO	NO	NO
BL2_VB2063A	Condition 2	Circulate time	60min, Stop AR2063	Stop	NO	NO	NO	NO	NO
BL2_VB2061B	Condition 1	Quiver time	2min, Start CR2061	Start	NO	NO	NO	NO	NO
BL2_VB2061B	Condition 2	Circulate time	10min, Stop CR2061	Stop	NO	NO	NO	NO	NO
BL2_VB2062B	Condition 1	Quiver time	4min, Start CR2062	Start	NO	NO	NO	NO	NO
BL2_VB2062B	Condition 2	Circulate time	10min, Stop CR2062	Stop	NO	NO	NO	NO	NO
BL2_VB2063B	Condition 1	Quiver time	6min, Start CR2063	Start	NO	NO	NO	NO	NO
BL2_VB2063B	Condition 2	Circulate time	30min, Stop CR2063	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL2_HT2061	Condition 1	Trip or stops or MFT signal	FN2001 / FN2003 / FN2002 Trip or stops or MFT signal, Stop HT2061	Stop	NO	NO	NO	NO	NO
BL2_HT2061	Condition 2	Ash hopper high level signal	LSH2061 Ash hopper high level signal Stop HT2061	Stop	NO	NO	NO	10min	NO
BL2_HT2062	Condition 1	Stop signal or MFT signal	FN2001 / FN2003 / FN2002 Stops signal or MFT signal Stop HT2062	Stop	NO	NO	NO	NO	NO
BL2_HT2062	Condition 2	Ash hopper high level signal	LSH2062 Ash hopper high level signal, Stop HT2062	Stop	NO	NO	NO	10min	NO
BL2_HT2063	Condition 1	Stop signal or MFT signal	FN2001 / FN2003 / FN2002 Stops signal or MFT signal, Stop HT2063	Stop	NO	NO	NO	NO	NO
BL2_HT2063	Condition 2	Ash hopper high level signal	LSH2063 Ash hopper high level signal, Stop HT2063	Stop	NO	NO	NO	10min	NO
BL2_FN2003	Condition 1	Trip or stops	FN2013 Trip or stops, Stop FN2003	Stop	NO	NO	NO	NO	NO
BL2_FN2003	Condition 2	No MFT signal	No MFT signal, Start FN2003	Start	NO	NO	NO	NO	NO
BL2_FN2003	Condition 3	No TT2003A HH: 80	No TT2003A HH: 80, Start FN2003	Start	NO	NO	NO	NO	NO
BL2_FN2003	Condition 4	No TT2003B HH: 80	No TT2003B HH: 80, Start FN2003	Start	NO	NO	NO	NO	NO
BL2_FN2011	Condition 1	Trip or stops	FN2003 Trip or stops, Stop FN2011	Stop	NO	NO	NO	NO	NO
BL2_FN2001	Condition 1	Trip or stops	FN2011 Trip or stops, Stop FN2001	Stop	NO	NO	NO	NO	NO
BL2_FN2001	Condition 2	No MFT signal	No MFT signal, Start FN2001	Start	NO	NO	NO	NO	NO
BL2_FN2001	Condition 3	No TT2001A HH: 80	No TT2001A HH: 80, Start FN2001	Start	NO	NO	NO	NO	NO
BL2_FN2001	Condition 4	No TT2001B HH: 80	No TT2001B HH: 80, Start FN2001	Start	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL2_INT8	Condition 1	Trip or stops	FN2001 Trip or stops, Stop FN2012, FN2014, FN2015, FN2004	Stop	NO	NO	NO	NO	NO
BL2_INT8	Condition 2	MV2001_ZSO signal ok	MV2001_ZSO signal ok, Start FN2014	Start	NO	NO	NO	NO	NO
BL2_INT8	Condition 3	MV2001_ZSO signal ok	MV2002_ZSO signal ok, Start FN2014	Start	NO	NO	NO	NO	NO
BL2_FN2002	Condition 1	Trip or stops	FN2012 Trip or stops, Stop FN2002	Stop	NO	NO	NO	NO	NO
BL2_FN2002	Condition 2	No MFT signal	No MFT signal, Start FN2002	Start	NO	NO	NO	NO	NO
BL2_FN2002	Condition 3	No TT2002A HH: 80	No TT2002A HH: 80, Start FN2002	Start	NO	NO	NO	NO	NO
BL2_FN2002	Condition 4	No TT2002B HH: 80	No TT2002B HH: 80, Start FN2002	Start	NO	NO	NO	NO	NO
BL2_BC2001	Condition 1	Trip or stops	FN2014 Trip or stops, Stop BC2001	Stop	NO	NO	NO	NO	NO
BL2_BC2001	Condition 2	No MFT signal	No MFT signal, Start BC2001	Start	NO	NO	NO	NO	NO
BL2_BC2001	Condition 3	No LSH2051 signal	No LSH1051 signal, Start BC2001	Start	NO	NO	NO	NO	NO
BL2_BC2001	Condition 4	No BC2001_AD signal	No BC2001_AD signal, Start BC2001	Start	NO	NO	NO	NO	NO
BL2_BC2002	Condition 1	Trip or stops	FN1015 Trip or stops, Stop BC2002	Stop	NO	NO	NO	NO	NO
BL2_BC2002	Condition 2	No MFT signal	No MFT signal, Start BC2002	Start	NO	NO	NO	NO	NO
BL2_BC2002	Condition 3	No LSH2052 signal	No LSH2052 signal, Start BC2002	Start	NO	NO	NO	NO	NO
BL2_BC2002	Condition 4	No BC2002_AD signal	No BC2002_AD signal, Start BC2002	Start	NO	NO	NO	NO	NO
BL2_INT9	Condition 1	Trip or stops	BC2001 Trip or stops, Stop RF232	Stop	NO	NO	NO	NO	NO
BL2_INT9	Condition 2	Trip or stops	BC2002 Trip or stops, Stop RF232	Stop	NO	NO	NO	NO	NO
BL2_INT9	Condition 3	PT230E H 0.4 MPa	PT230E H 0.4 Mpa, Stop RF232	Stop	NO	NO	NO	15s	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BL2_INT9	Condition 4	No MFT signal	No MFT signal, Start RF232	Start	NO	NO	NO	NO	NO
BLO_INT1	Condition 1	Deaerator water level HH	LT020 HH: 2200mm, Stop PU005A & PU005B	Stop	NO	NO	NO	NO	NO
BLO_INT2	Condition 1	Level LT011 $\leq$ 100mm	LT011 $\leq$ 100mm, Stop PU11 & PU012	Stop	NO	NO	NO	NO	NO
BL1_BI1001ON2	Condition 1	Temperature TT1030 $\geq$ 900	TT1030 $\geq$ 900, Stop BI1001_ON2(OFF)	Stop	NO	NO	NO	NO	NO
BL1_BI2001ON2	Condition 1	Temperature TT2030 $\geq$ 900	TT2030 $\geq$ 900, Stop BI2001_ON2(OFF)	Stop	NO	NO	NO	NO	NO
	Condition 1	Temperature TT1030 $\geq$ 900	TT1030 $\geq$ 900, MV011 - Full open output 100%	Open	NO	NO	NO	NO	NO
	Condition 2	Temperature TT2030 $\geq$ 900	TT2030 $\geq$ 900, MV011 - Full open output 100%	Open	NO	NO	NO	NO	NO
BLO_PU0012	Condition 1	PU011 Trip or stop	PU011 Trip or stop, Start PU012	Start	NO	NO	NO	NO	NO
BLO_PU0011	Condition 1	PU012 Trip or stop	PU012 Trip or stop, Start PU011	Start	NO	NO	NO	NO	NO
BLO_PU0081A	Condition 1	Ammonia tank level LL	LT081 LL: 100, Stop PU081A	Stop	NO	NO	NO	NO	NO
BLO_PU0082A	Condition 1	Sodium sulfite tank level LL	LT082 LL: 100, Stop PU082A	Stop	NO	NO	NO	NO	NO
BLO_PU0083A	Condition 1	Phosphate tank level LL	LT083 LL: 100, Stop PU083A	Stop	NO	NO	NO	NO	NO
BLO_PU0084A	Condition 1	Phosphate tank level LL	LT084 LL: 100, Stop PU084A	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BLO_RL0230	Condition 1	DC230 Trip or stops	DC0230 Trip or stops, Stop RF230	Stop	NO	NO	NO	NO	NO
BLO_RL0230	Condition 2	LSH231 High signal	LSH231 High signal, Stop RF230	Stop	NO	NO	NO	30 s	NO
BLO_FN0030	Condition 1	DC230 Trip or stops	DC0230 Trip or stops, Stop FN230	Stop	NO	NO	NO	NO	NO
BLO_HE0030	Condition 1	FN230 Trip or stops	FN230 Trip or stops, Stop HE230	Stop	NO	NO	NO	NO	NO
BLO_HE0030	Condition 2	TT230 H: 90	TT230 H: 90, Stop HE230	Stop	NO	NO	NO	NO	NO
BLO_HE0030	Condition 3	TT230 L: 50	TT230 L: 50, Start HE230	Start	NO	NO	NO	NO	NO
BLO_RL0231	Condition 1	Boiler 1 Limestone conveying pressure	PT230D pressure H: 0.4MPa, Stop RF231	Stop	NO	NO	NO	15 s	NO
BLO_RL0232	Condition 1	Boiler 2 Limestone conveying pressure	PT230E pressure H: 0.4MPa, Stop RF232	Stop	NO	NO	NO	15 s	NO
BLO_SC0051A	Condition 1	BC1001 trip or stops	BC1001 trip or stops, SC051A Stop	Stop	NO	NO	NO	NO	NO
BLO_SC0051A	Condition 2	Level Switch Signal	LSL051A signal, SC051A Stop	Stop	NO	NO	NO	1 m	NO
BLO_SC0051A	Condition 3	SC051B trip or stops	SC051B trip or stops, SC051A Stop	Stop	NO	NO	NO	NO	NO
BLO_RL0051A	Condition 1	SC051A trip or stops	SC051A trip or stops, RL051A Stop	Stop	NO	NO	NO	NO	NO
BLO_RL0051A	Condition 2	Level Switch Signal	LSL051A signal, RL051A Stop	Stop	NO	NO	NO	NO	NO
BLO_EL0051A	Condition 1	RL051A trip or stops	RL051A trip or stops, BE051A Stop	Stop	NO	NO	NO	NO	NO
BLO_EL0051A	Condition 2	Level Switch Signal	LSL051A signal, BE051A Stop	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BLO_SC0051B	Condition 1	BC2001 trip or stops	BC2001 trip or stops, SC051B Stop	Stop	NO	NO	NO	NO	NO
BLO_SC0051B	Condition 2	Level Switch Signal	LSL051A signal, SC051B Stop	Stop	NO	NO	NO	2 m	NO
	Condition 1	No speed monitoring signal after 1 minute	No speed monitoring signal, BC075 Stop	Stop	NO	NO	NO	NO	NO
	Condition 2	No anti-deviation device signal	No anti-deviation device signal, Start BC075	Start	NO	NO	NO	NO	NO
	Condition 3	No 2 way accident rope switch signal	No 2 way accident rope switch signal, Start BC075	Start	NO	NO	NO	NO	NO
BLO_INT3	Condition 1	BC075 Trip or stops	BC075 Trip or stops, FN1016, FN2016 Stop	Stop	NO	NO	NO	NO	NO
BL1_SC1001	Condition 1	FN1016 Trip or stops	FN1016 Trip or stops, SC1001 Stop	Stop	NO	NO	NO	NO	NO
BL1_SC1001	Condition 2	Temperature Signal	TT1025 HH: 75, SC1001 Stop	Stop	NO	NO	NO	NO	NO
BL1_SC1001	Condition 3	FSL1025 Signal	FSL1025 Signal, SC1001 Stop	Stop	NO	NO	NO	NO	NO
BL2_SC2001	Condition 1	FN2016 Trip or stops	FN2016 Trip or stops, SC2001 Stop	Stop	NO	NO	NO	NO	NO
BL2_SC2001	Condition 2	Temperature Signal	TT1025 HH: 75, SC2001 Stop	Stop	NO	NO	NO	NO	NO
BL2_SC2001	Condition 3	FSL1025 Signal	FSL1025 Signal, SC2001 Stop	Stop	NO	NO	NO	NO	NO
BLO_INT4	Condition 1	Remote control signal ok	Remote control signal ok Open DC051/DC052	Open	NO	NO	NO	NO	NO
	Condition 1	MV051/MV052 push down control fall-position signal			NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BLO_BC054	Condition 1	No anti deviation device signal	No anti deviation device signal, Start BC054	Start	NO	NO	NO	NO	NO
BLO_BC054	Condition 2	No two way accident rope switch signal	No two way accident rope switch signal, Start BC054	Start	NO	NO	NO	NO	NO
BLO_BC054	Condition 3	No speed monitoring device signal after 1 minute	No speed monitoring device signal, Stop BC054	Stop	NO	NO	NO	NO	NO
BLO_BC053	Condition 1	No anti deviation device signal	No anti deviation device signal, Start BC053	Start	NO	NO	NO	NO	NO
BLO_BC053	Condition 2	No two way accident rope switch signal	No two way accident rope switch signal, Start BC053	Start	NO	NO	NO	NO	NO
BLO_BC053	Condition 3	No speed monitoring device signal after 1 minute	No speed monitoring device signal, Stop BC053	Stop	NO	NO	NO	NO	NO
BLO_BC053	Condition 4	BC054 Trip or stops	BC054 Trip or stops, Stop BC053	Stop	NO	NO	NO	NO	NO
BLO_INT5	Condition 1	Speed alarm	Speed alarm, Stop CR051A & CR051B	Stop	NO	NO	NO	NO	NO
BLO_INT5	Condition 2	Flash back and soft starter alarm	Flash back and soft starter alarm, Stop CR051A & CR051B	Stop	NO	NO	NO	NO	NO
BLO_INT5	Condition 3	Temperature/over current alarm	Temperature/over current alarm, Stop CR051A & CR051B	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BLO_INT5	Condition 4	Motor protector alarm	Motor protector alarm, Stop CR051A & CR051B	Stop	NO	NO	NO	NO	NO
BLO_INT5	Condition 5	BC053 trip or stop	BC053 trip or stop, Stop CR051A & CR051B	Stop	NO	NO	NO	NO	NO
BLO_INT5	Condition 6	Remote control signal ok	Remote control signal ok, Start CR051A & CR051B	Start	NO	NO	NO	NO	NO
BLO_INT5	Condition 7	Remote power signal ok	Remote power signal ok, Start CR051A & CR051B	Start	NO	NO	NO	NO	NO
BLO_DET0051	Condition 1	CR051A Trip or stops	CR051A Trip or stops, Stop MD051	Stop	NO	NO	NO	NO	NO
BLO_DET0051	Condition 2	CR051B Trip or stops	CR051B Trip or stops, Stop MD051	Stop	NO	NO	NO	NO	NO
BLO_BC052	Condition 1	MD051 Trip or stops	MD051 Trip or stops, Stop BC052	Stop	NO	NO	NO	NO	NO
BLO_BC052	Condition 2	No speed monitoring signal after 1 minute	No speed monitoring signal after 1 minute, Stop BC052	Stop	NO	NO	NO	NO	NO
BLO_BC052	Condition 3	No anti-deviation device signal	No anti-deviation device signal, Start BC052	Start	NO	NO	NO	NO	NO
BLO_BC052	Condition 4	No 2 way accident rope switch signal	No 2 way accident rope switch signal, Start BC052	Start	NO	NO	NO	NO	NO
BLO_MA0051	Condition 1	BC052 Trip or stops	BC052 Trip or stops, Stop MS051	Stop	NO	NO	NO	NO	NO
BLO_BC051	Condition 1	MS051 Trip or stops	MS051 Trip or stops, Stop BC051	Stop	NO	NO	NO	NO	NO

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Module	Name	Description	Expression	Interlock State	Disable	Higher Managed	Reset Required	Delay Off	Delay On
BLO_BC051	Condition 2	No speed monitoring signal after 1 minute	No speed monitoring signal after 1 minute, Stop BC051	Stop	NO	NO	NO	NO	NO
BLO_BC051	Condition 3	No anti-deviation device signal	No anti-deviation device signal, Start BC051	Start	NO	NO	NO	NO	NO
BLO_BC051	Condition 4	No 2 way accident rope switch signal	No 2 way accident rope switch signal, Start BC051	Start	NO	NO	NO	NO	NO
BLO_VB0051	Condition 1	BC051 Trip or stops	BC051 Trip or stops, Stop VB051	Stop	NO	NO	NO	NO	NO

Table 33: List of interlocks and modules

6 MODULE PERMISSIVE

6.1 Functional Description

The section below explains the permissive configurations

Module	Name	Description	Expression	Disable	Delay Off	Delay On
BL1_SC1001	Condition 1	BC054 running start permissive	Start SC1001	NO	NO	NO
BL0_PU0001	Condition 1	FN001 running start permissive	Start PU001	NO	NO	NO
BL0_PU0002	Condition 1	FN002 running start permissive	Start PU002	NO	NO	NO
BL0_RL0230	Condition 1	DC230 running start permissive	Start RF230	NO	NO	NO
BL0_FN0030	Condition 1	DC230 running start permissive	Start FN230	NO	NO	NO
BL0_HE0030	Condition 1	FN230 running start permissive	Start HE230	NO	NO	NO
BL2_SC2001	Condition 1	BC054 running start permissive	Start SC2001	NO	NO	NO
BL0_PU0003	Condition 1	FN003 running start permissive	Start PU003	NO	NO	NO
BL0_PU0004	Condition 1	FN004 running start permissive	Start PU004	NO	NO	NO
BL0_SC0051A	Condition 1	BC1001 running start permissive	Start SC051A	NO	NO	NO

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Module	Name	Description	Expression	Disable	Delay Off	Delay On
BL0_RL0051A	Condition 1	SC051A running start permissive	Start RL051A	NO	NO	NO
BL0_EL0051A	Condition 1	RL051A running start permissive	Start BE051A	NO	NO	NO
BL0_SC0051B	Condition 1	BC2001 running start permissive	Start SC051B	NO	NO	NO
BL_PERM1	Condition 1	BC075 running start permissive	Start FN1016, FN2016	NO	NO	NO
BL1_SC1001	Condition 2	FN1016 running start permissive	Start SC1001	NO	NO	NO
BL2_SC2001	Condition 2	FN2016 running start permissive	Start SC2001	NO	NO	NO
BL1_FN1003	Condition 1	FN1013 running start permissive	Start FN1003	NO	NO	NO
BL1_FN1011	Condition 1	FN1003 running start permissive	Start FN1011	NO	NO	NO
BL1_FN1001	Condition 1	FN1011 running start permissive	Start FN1001	NO	NO	NO
BL_PERM2	Condition 1	FN1001 running start permissive	Start FN1012, FN1014, FN1015, FN1004	NO	NO	NO
BL1_FN1002	Condition 1	FN1012 running start permissive	Start FN1002	NO	NO	NO
BL1_BC1001	Condition 1	FN1014 running start permissive	Start BC1001	NO	NO	NO
BL1_BC1002	Condition 1	FN1015 running start permissive	Start BC1002	NO	NO	NO
BL0_RL0231	Condition 1	BC1001 running start permissive	Start RF231	NO	NO	NO

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Module	Name	Description	Expression	Disable	Delay Off	Delay On
BL0_RL0231	Condition 2	BC1002 running start permissive	Start RF231	NO	NO	NO
BL2_FN2003	Condition 1	FN2013 running start permissive	Start FN2003	NO	NO	NO
BL2_FN2011	Condition 1	FN2003 running start permissive	Start FN2011	NO	NO	NO
BL2_FN2001	Condition 1	FN2011 running start permissive	Start FN2001	NO	NO	NO
BL_PERM3	Condition 1	FN2001 running start permissive	Start FN2012, FN2014, FN2015, FN2004	NO	NO	NO
BL2_FN2002	Condition 1	FN2012 running start permissive	Start FN2002	NO	NO	NO
BL2_BC2001	Condition 1	FN2014 running start permissive	Start BC2001	NO	NO	NO
BL2_BC2002	Condition 1	FN2015 running start permissive	Start BC2002	NO	NO	NO
BL0_RL0232	Condition 1	BC2001 running start permissive	Start RF232	NO	NO	NO
BL0_RL0232	Condition 2	BC2002 running start permissive	Start RF232	NO	NO	NO
BL0_BC054	Condition 1	MV051 start permissive	Start BC054	NO	NO	NO
BL0_BC054	Condition 2	MV052 start permissive	Start BC054	NO	NO	NO
BL0_BC053	Condition 1	BC054 running start permissive	Start BC053	NO	NO	NO
BL0_CR051A/B	Condition 1	BC053 running start permissive	Start CR051A or CR051B	NO	NO	NO

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Module	Name	Description	Expression	Disable	Delay Off	Delay On
BL0_DET0051	Condition 1	CR051A running start permissive	Start MD051	NO	NO	NO
BL0_DET0051	Condition 2	CR051B running start permissive	Start MD051	NO	NO	NO
BL0_BC052	Condition 1	MD051 running start permissive	Start BC052	NO	NO	NO
BL0_MA0051	Condition 1	BC052 running start permissive	Start MS051	NO	NO	NO
BL0_BC051	Condition 1	MS051 running start permissive	Start BC051	NO	NO	NO
BL0_VB0051	Condition 1	BC051 running start permissive	Start VB051	NO	NO	NO

Table 34: List of interlocks and modules